BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, MARDAN

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: MRD-XI-21(A) [INCOMPLETE: Objective not found]

Retyped faithfully from the original board paper — same questions, no changes.

Section – B

Time Allowed: 2:40 Hours | **Maximum Marks:** 50

Note: [INCOMPLETE PAPER: Section-A (Objective) not found on any source — subjective only.]

QII. Write short answers of any ten (10) of the following parts. All parts carry equal marks.

1. If $Z_1 = 2 + i$ and $Z_2 = 1 - i$ then verify that $Z_1 + Z_2 = Z_2 + Z_1$
2. If $A = [1\ 2\ 0; 3\ -1\ 4]$, show that $(A^t)^t = A$
3. Find α, so that $|\alpha i + (\alpha + 1)j + 2k| = 3$
4. If $a_{10} = l$, $a_{13} = m$, $a_{16} = n$; show that $ln = m^2$
5. Find the sum $1^2 + 3^2 + 5^2 + 7^2 + \ldots\ldots + 99^2$
6. Prove that $(n+5)!/(n+3)! = n^2 + 9n + 20$
7. Show by Mathematical Induction that $(10^{n+1} - 9n - 10)/81$ is an integer.
8. Sketch the graph of function $f(x) = (x - 1)(x - 3)$
9. Show that $\sin(\alpha+\beta)/(\cos\alpha\cos\beta) = \tan\alpha + \tan\beta$
10. Solve the triangle with dimensions $a = 209$, $b = 120$, $c = 241$
11. Find the period of $y = \cos 2\theta$
12. Find the ranks of $A = [3\ 1\ -4; 0\ 2\ 1; 1\ -1\ -2]$
13. Find the number of terms in the arithmetic progression 6,9,12,.....,78.

Section – C

Maximum Marks: 30

Note: Answer any three (3) questions. All questions carry equal marks.

QIII.

(a) To Solve by Cramer's rule, if $x - y + 2z = 10$, $2x + y - 2z = -4$, $3x + y + z = 7$

(b) Find the area of a parallelogram whose diagonals are. $\vec{a} = 3i + 2j - 2k$, $\vec{b} = i - 3j + 4k$

QIV.

(a) Prove that ${}^{n}C_{r} + {}^{n}C_{r-1} = {}^{n+1}C_{r}$

(b) Find the middle term in the expansion of $(3x + x^2/2)^9$

QV.

(a) Prove that $(\sin\alpha + \sin9\alpha)/(\cos\alpha + \cos9\alpha) = \tan5\alpha$

(b) Show that $abc(\sin\alpha + \sin\beta + \sin\gamma) = 4\Delta s$

QVI.

(a) Show that the sum of the first n positive odd integers is n^2.

(b) Find the graph of the function $y = 2\sin x$; $0 \leq x \leq 2\pi$